

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy conservation: transport, buildings, industry - Set 1

Mark scheme

AQA spec 3.3.4.2; Energy resources; textbook pp. 225-235

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for the correct answer.</i> • $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%) (1)	1
01.2	<i>1 mark for each point.</i> • Although the larger vehicle uses more energy in total, it carries many more people / a larger load (1) • So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance (1)	2
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Flexibility of routes / destinations and door-to-door convenience (1) • Time / speed and reliability where public transport is infrequent (1) • Low fuel tax / cheap fuel (1) • Carrying goods; comfort; status (1)	3
01.4	<i>1 mark for each point.</i> • Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey (1) • So more journeys shift from cars to bulk transport / cycling (1) • ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely (1) • Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships (2) - Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption (2) - Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again (2)	6
02.2	<i>1 mark for each point.</i> - Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) (1) - Steel wheels do not deform so rolling resistance is very low (1)	2
02.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel (1) - Better temperature control (thermostatic fan) (1) - Electronic ignition control / regular servicing to keep spark timing correct (1) - Turbochargers supplying more oxygen (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
03.1	<i>1 mark for each point.</i> - The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) (1) - Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) (1) - Lighter designs using less material; longer vehicle life spreading the embodied energy (1)	3
03.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Use of recyclable materials wherever possible (1) - Easy identification of components and their composition (code stamps) (1) - Easy dismantling and separation of components / materials (1) - Reusable components for new vehicles (1) - Compostable materials for parts that cannot be recycled (1)	4
03.3	<i>1 mark for each point.</i> - At low speed a large proportion of the energy just runs the engine rather than moving the car (1) - At high speed energy is wasted overcoming air resistance (which rises with the square of speed); optimum for many cars about 56 mph (1)	2
03.4	<i>1 mark.</i> Automatic stop / start systems cutting the engine when stationary; smooth driving in higher gears avoiding sudden braking / acceleration (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
04.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Single: $5.8 \times 20 \times 12 = 1392 \text{ W}$ (1) • Argon: $1.6 \times 20 \times 12 = 384 \text{ W}$ (1) • Reduction = 1008 W (accept 1010) (1)	3
04.2	<i>1 mark for each point.</i> • A wider gap of trapped gas reduces conduction because gas is a poor conductor (1) • If the gap is too wide convection currents start in the gas, carrying heat across; the optimum depends on the gas and window size (1)	2
04.3	<i>1 mark for each point.</i> • Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses (1) • Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap (1)	2
04.4	<i>1 mark for each point.</i> • Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm (1) • Used where there is not room for thick gas-filled multiple glazing (eg historic buildings) (1)	2
04.5	<i>1 mark.</i> • Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
05.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Orientation: south-facing windows (northern hemisphere) maximise passive solar gains; rooms needing warmth on the sunny side; fewer / smaller windows on the north side where losses are greatest; solar screens / brise soleil prevent summer overheating (2) • Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost (2) • Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced (2)	6
05.2	<i>1 mark for each point.</i> • Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy (1) • Limecrete has about half the embodied energy of concrete; rammed earth, straw bales and lime mortar are lower still (1)	2
05.3	<i>1 mark for each point.</i> • Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area (1) • Heat transfers from the outgoing to the incoming air so fresh air enters already warmed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room (1) • Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated (1)	2
06.2	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> • LEDs convert a much larger proportion of electricity to light rather than heat (up to 2) • Washing machines with cold / low-temperature cycles and fast spins (less drying); dishwashers using less water so less heating; cookers with double / triple glazed doors; fridges with linear compressors; LED screens using half the power of plasma; germanium transistors (up to 2)	3
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Turning off lights / appliances rather than standby (1) • Turning thermostats and radiator valves down / off in unused rooms (1) • Only heating the water needed (kettle) (1) • Showering instead of bathing; eco-cycles (1) • Choosing smaller vehicles / less travel (1)	3
06.4	<i>1 mark for each point.</i> • Stored hot water loses heat continuously through the tank walls and pipes (1) • Heating as needed (or just before) avoids these standing losses; if storage is needed the tank should be well insulated (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Conventional: $40 / 100 = 40\%$ (1) • CHP: $(35 + 45) / 100 = 80\%$ (1)	2
07.2	<i>1 mark for each point.</i> • Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature (1) • Hotter water is more useful for district heating of homes, fish farms, greenhouses (1)	2
07.3	<i>1 mark.</i> • Hot water loses heat and needs pumping energy over distance; heat cannot be transported as far as electricity (1)	1
07.4	<i>1 mark for each point.</i> • Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly (1) • Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred (1) • Slower flow allows more time for heat to transfer (1)	3
07.5	<i>1 mark for each point.</i> • One large tank has a lower surface area : volume ratio than many small ones (1) • A sphere has the smallest surface area for any volume; less surface means less heat loss (with insulation of low thermal conductivity) (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
08.1	<i>1 mark for each point.</i> • Molten iron from the blast furnace is converted directly to steel on the same site (1) • So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg) (1)	2
08.2	<i>1 mark for each point.</i> • Collecting and processing small dispersed quantities can use more energy in transport than is saved (1) • Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times (1)	2
08.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling (2) • ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided (2)	4
08.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Proximity to the grid / demand to reduce transmission losses and cable costs (1) • Access to cooling water for thermal stations (1) • Resource availability (wind, tidal range, fuel deliveries) (1) • Land availability and planning / public acceptance; stable geology (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	